

QUESTIONS

Q1
Prime Number Checker
20 marks

Q2
Factorial Calculation
20 marks

Q3
Largest Element in an Array
20 marks

Q4
Palindrome String Checker
20 marks

Q5
Simple Interest Calculator
20 marks

5/5 answered

Q1 Prime Number Checker

20 marks

Write pseudocode for a program that accepts an integer from the user and determines whether it is a prime number. The pseudocode should include a loop that checks divisibility from 2 up to the square root of the number (or up to $n-1$), and should clearly state the decision logic used to conclude whether the number is prime or not.

Key Elements Expected: input step, loop with divisibility check, boolean/flag tracking, final decision output.

Your Answer

```

START
1.Declare Integer n,i
2.Declare Boolean isprime
3.Display "Enter a number:"
4.Read n
5.If n<=1 Then
Display n,"is not a prime number"
STOP
End If
6.Set isprime←True
7.For i←2 to sqrt(n) Do
If n MOD i=0 Then
isprime←False
Exit For
End If
End For
8.If isprime=True Then
Display n,"is a prime number"
Else
Display n,"is not a prime number"
End If
STOP
    
```

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Q2 Factorial Calculation

20 marks

Write pseudocode for a program that accepts a non-negative integer n from the user and calculates its factorial ($n!$) using an iterative loop (not recursion). Clearly show the initialization of the result variable, the loop structure, and the final output step.

Key Elements Expected: variable initialization, loop from 1 to n , running product update, output step.

Your Answer

```
START
1.Declare Integer n,i
2.Declare Long Integer fact
3.Display "Enter a non-negative integer:"
4.Read n
5.If n<0 Then
Display "Factorial is not defined for negative numbers"
STOP
End If
6.fact=1
7.For i=1 to n Do
fact=fact*i
End For
8.Display "Factorial of",n,"=",fact
STOP
```

Save Answer

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Q3 Largest Element in an Array

20 marks

Write pseudocode for a program that accepts an array of n integers and determines the largest element in the array. The pseudocode should show the initialization of a variable to track the maximum value, the loop used to traverse the array, and the comparison logic used to update the maximum value when a larger element is found.

Key Elements Expected: max-variable initialization, traversal loop, comparison and update logic, final output.

Your Answer

```
START
1.Declare Integer n,i
2.Declare Integer array[n]
3.Declare Integer max
4.Display "Enter the number of elements:"
5.Read n
6.Display "Enter the array elements:"
7.For i=0 to n-1 Do
  Read array[i]
End For
8.max =array[0]
9.For i=1 to n-1 Do
  If array[i]>max Then
    max=array[i]
  End If
End For
10.Display "the largest element is:",max
STOP
```

 Save Answer

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5/5 answered

Q4 Palindrome String Checker

20 marks

Write pseudocode for a program that accepts a string from the user and determines whether it is a palindrome (reads the same forwards and backwards). The pseudocode should show how the string is compared character by character (or compared with its reverse), and should clearly state the final decision output.

Key Elements Expected: input step, comparison logic (two-pointer or reversal approach), final decision output.

Your Answer

```

START
1.Declare String str
2.Declare Integer left,right
3.Declare Boolean isPalindrome
4.Display "Enter a string:"
5.Read str
6.left=0
7.right=Length(str)-1
8.isPalindrome=True
9.While left<right Do
If str[left] != str[right] Then
isPalindrome=False
Exit While
End If
left=left+1
right=right-1
End While
10.If isPalindrome=True Then
Display "the string is a palindrome"
Else
Display "the string is not a palindrome"
End If
STOP
    
```

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Q5 Simple Interest Calculator

20 marks

Write pseudocode using the IPO (Input, Process, Output) structure to calculate the simple interest on a given principal amount. The pseudocode should clearly show the INPUT section (principal, rate of interest, time in years), the PROCESS section (applying the simple interest formula), and the OUTPUT section (displaying the calculated interest).

Key Elements Expected: INPUT section listing principal, rate, time; PROCESS section showing $SI = (P \times R \times T) / 100$; OUTPUT section displaying the result.

Your Answer

```

START
1.Declare Float P,R,T,SI
INPUT
2.Display "Enter principal amount:"
3.Read P
4.Display "Enter Rate of Interest:"
5.Read R
6.Display "Enter Time (in years):"
7.Read T
PROCESS
8.SI=(P*R*T)/100
OUTPUT
9.Display "Simple Interest=",SI
STOP
  
```

 Save Answer